

Technology Stack

Date	10 July 2026
Team ID	xxxxxx
Project Name	Raising Waters
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript	<i>Chosen to build a lightweight, responsive, and clean user interface for the Flask web application, allowing disaster coordinators to view risk alerts efficiently on any device.</i>

2	Backend / Server-Side	<i>Python / Flask</i>	<i>Flask provides a lightweight, minimalist framework perfect for serving our prediction models and handling data ingestion routing with very low latency.</i>
3	Database / Data Storage	<i>Pickle (.pkl) & Local CSV / SQLite</i>	<i>Used for storing trained model weights seamlessly from Python, alongside structured logs of historical predictions for fast verification and auditing.</i>
4	Cloud / Hosting / Deployment	<i>Localhost Deployment</i>	<i>Setting up the local server environment ensures zero-cost deployment and immediate local testing of real-time data ingestion pipelines during the development phase.</i>
5	Version Control & CI/CD	<i>GitHub</i>	<i>[Chosen for seamless code version control, structured tracking of project updates, and hosting the repository for easy sharing and backup.</i>

6	Third-Party APIs / Other Tools	<i>OpenWeatherMap API (or standard Weather API)</i>	<i>Enables the application to automatically pull live, real-time meteorological metrics, rainfall depth, and cloud coverage data to feed directly into the ML inference system.</i>